RISHI MADHUR KANAJAM

Full Stack AI Engineer
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SUMMARY

Full stack AI engineer who builds AI systems that show how far they can be trusted: retrieval pipelines with evaluation harnesses, LLM benchmarks, and health-data infrastructure with compliance checks built in. Works end to end across Python/FastAPI services, React/Next.js interfaces and cloud deployment on AWS, Azure and GCP. Maintainer of Krama Core, an open-source FHIR R4 SDK on PyPI. Full Australian working rights; no sponsorship required.

TECHNICAL SKILLS

AI & ML: Generative AI, LLM agents and agentic workflows, RAG (hybrid BM25 + dense, re-ranking), LangChain, LangGraph, Semantic Kernel, prompt engineering, structured outputs, LLM-as-judge evaluation; Anthropic, OpenAI, Gemini and Groq APIs; PyTorch, Hugging Face, YOLOv8, TensorRT

Frontend: React, Next.js, Angular, TypeScript / JavaScript, HTML5, CSS3, Tailwind CSS, React Native

Backend: Python (primary), FastAPI, Flask, Node.js / Express, REST APIs, WebSockets, Pydantic, Alembic, Redis, C++

Data & Analytics: SQL, ETL/ELT, Pandas, NumPy, Spark, Airflow, Snowflake; FHIR R4, ABDM

Cloud: AWS (Certified AI Practitioner), Azure (AI Language, data services), GCP (Cloud Run, Vertex AI), Firebase, Vercel

DevOps & MLOps: Docker, GitHub Actions, CI/CD, pytest, ruff, bandit, pip-audit, INT8 quantisation, pruning, inference profiling

Databases: PostgreSQL, MongoDB Atlas (Vector Search), ChromaDB, Firestore, SQLite

PROFESSIONAL EXPERIENCE

Academic Tutor, Data Structures & Algorithms (COMP2123/COMP9123) **01/2024 – Present**

University of Sydney

- Teach tutorials and labs on trees, priority queues, graph traversals, dynamic programming and complexity analysis to 50+ undergraduate and postgraduate students per semester.
- Grade assignments with detailed written feedback, improve lab materials and test cases, and built an interactive React platform for visualising data structures from the concepts students struggle with most.

Systems Engineer – Data Team (Part-time) **04/2025 – 03/2026**

St Vincent de Paul Society NSW (Vinnies)

- Built Python scripts to automate data extraction and cleansing, reducing manual effort in the team's ETL processes.
- Supported data integration workflows across SAP, Workday and Azure data services: mapped fields, validated outputs, troubleshot pipeline issues, and documented data flows and transformation logic.

Computer Vision Engineer (University Capstone) **08/2024 – 12/2024**

Ocius Technology

- Worked with Ocius on a real-time maritime object-detection pipeline (YOLOv8) for its autonomous Bluebottle surface vessels, as part of a four-person capstone team.
- Owned model optimisation: INT8 quantisation, pruning, TensorRT, inference batching and profiling, delivering the optimised model behind a team-wide ~9× throughput gain on NVIDIA Jetson edge hardware; containerised the pipeline with Docker.

Frontend Developer (Contract) **03/2023 – 07/2023**

HRVITE Services Pvt Ltd

- Re-engineered Angular UI components and interaction flows for an HR SaaS platform, contributing to a measured ~15% increase in user retention.
- Built backend REST APIs for the platform and the data-driven frontend components that consumed them, with state management for nested HR datasets.
- Wrote unit tests and took part in code reviews to reduce regressions.

Software Engineer (Trainee) **06/2022 – 01/2023**

Revidd

- Moved from a two-month traineeship into full engineering responsibilities at an early-stage startup, working largely independently.
- Built the frontend for a new platform, maintained WordPress sites, and worked on CI/CD pipelines for a microservices architecture.

KEY PROJECTS

Krama Core – Open-Source FHIR & Compliance SDK | *Maintainer, Nirvya Labs* | Source | PyPI

- Python SDK for India's ABDM health network: ABHA identity, consent and data-exchange flows, FHIR R4 bundle builders, ECDH X25519 + AES-GCM encryption, gateway retries and circuit breakers, and clinical templates across 12 domains.
- Compliance engine returns blockers, not warnings, with country rule packs (India, Australia, US, UK). v1.0.0-alpha.3 on PyPI, Apache 2.0, 128 tests, CI on Python 3.10–3.12.

PathFinder – AI Triage & Referral Platform | *NGM Group Hackathon, Idea Disruptor Award, built with a teammate* | Source

- 24/7 intake chatbot, QR-based referral portal and staff dashboard for a regional suicide-prevention network; FastAPI, Azure AI Language, React.
- Rule-based risk engine routes referrals across 14 programs, backed by a five-layer escalation protocol (email, push, SMS), 35+ documented edge cases, and crisis numbers hardcoded in static HTML so they survive a backend outage.

PDF-Rag – Hybrid Retrieval with Evaluation | Source

- BM25 and dense search merged with Reciprocal Rank Fusion, cross-encoder re-ranking, sentence-aware chunking and conversational memory, plus an LLM-as-judge harness scoring faithfulness, relevancy and context quality.

Pinch Recovery Engine – Failed-Payment Recovery | *Pinch 50K Hackathon* | Source

- Classifies direct-debit dishonour codes and applies a strategy per failure class: never retries hard failures, schedules only on Australian business days, and logs human-readable reasoning for every decision. FastAPI, PostgreSQL, Alembic, Docker; idempotent webhooks; runs in mock mode or against the live Pinch sandbox.

ChronoLens – Cultural Research Workspace | *OpenAI Codex Hackathon Sydney, solo* | Source

- Built in about five hours: one query fans out to ten analysis views and live archive APIs, and every claim is labelled fact, context, hypothesis or needs-review through to the exported deck. Next.js 14, TypeScript, GPT-4.1.

MedVault – Multi-Tenant Clinical AI Platform | Source

- Gemini multimodal decision support with mandatory physician review, IoT cold-chain monitoring, offline-first records via IndexedDB, and tenant isolation enforced by Firebase custom claims and Firestore rules. Next.js 15, TypeScript, Cloud Run.

DeceptionArena – LLM Deception Benchmark | *In progress, targeting arXiv*

- Five social deduction games as pure state machines behind one engine interface (1,000/1,000 randomised stress tests), Deception ELO ratings, structured decision traces, aiohttp WebSocket server and React dashboard; 15 models across 5 providers wired in.

EDUCATION & CERTIFICATIONS

Master of IT & IT Management (Data Analytics & Management) – University of Sydney 2023 – 2025
B.Tech (Honours), Computer Science and Engineering – GMR Institute of Technology 2019 – 2023
AWS Certified AI Practitioner (2025) | McKinsey Forward Program (2026)